TOWARDS GLOBAL GEOMETRY PRESERVATION IN TEXT EMBEDDING DISTILLATION

Anonymous authors
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ABSTRACT

A text embedding model encodes relational knowledge in the global geometry it induces over a corpus. Relational knowledge distillation (KD) offers a natural way to transfer this knowledge, but existing objectives typically construct relations only among examples that share a mini-batch. Random batch co-occurrence rather than teacher relevance therefore selects the supervision support, leaving most corpus-level teacher signal outside the realized objective. We characterize this target-agnostic exposure and derive a finite-epoch bound that separates selected-support optimization error from missed teacher mass in global geometry distortion. Motivated by this analysis, we introduce **RIPPLE**, which uses a cached teacher graph to select corpus relations while retaining mini-batches only as computational units. RIPPLE combines multi-hop graph targets, mass-aware candidate selection, and explicit support-mass calibration; it requires neither task labels nor online teacher inference and adds no inference-time cost. Across nine datasets and three teacher–student pairs, RIPPLE obtains the best mean score among the evaluated compact students in all three settings, with the strongest and most consistent gains on semantic textual similarity.

1 INTRODUCTION

Text embedding models map examples into a space whose global arrangement supports similarity search, clustering, and transfer across tasks. A teacher therefore contains knowledge not only in individual vectors, but also in the relations it induces among corpus examples. Prior work has successfully distilled pairwise similarities, relative distances and angles, conditional affinity distributions, and topological summaries (Tung & Mori, 2019; Park et al., 2019; Passalis et al., 2021; Kim et al., 2023; 2024). These results motivate the transfer idea studied here: use the teacher's corpus-level geometry as relational knowledge for a compact student.

Relational KD realizes this idea by aligning teacher and student similarities, but commonly constructs its pair matrices or conditional distributions from the current mini-batch (Tung & Mori, 2019; Park et al., 2019; Passalis et al., 2021; Qian et al., 2022; Gou et al., 2025). Reusing encoded examples makes this efficient, yet also lets the data loader select relational support. A global relation is defined by teacher geometry; a batch-local relation is selected by random co-occurrence. Thus the loss is relational, but the teacher does not choose the relations.

We quantify this limitation through *exposed teacher mass*: the probability mass that a global teacher target assigns to the relations observed during training. Our analysis shows that mini-batch coverage is governed by random co-occurrence rather than teacher relevance. Teacher-proportional selection instead adapts to how concentrated the teacher's relational signal is. This gap makes support selection a central design choice rather than a minor implementation detail.

The limitation is therefore not mini-batch computation, but mini-batch relation selection. **RIPPLE** follows the opposite design principle: *teacher geometry selects the relations; the mini-batch only computes them.* A cached teacher graph replaces random co-occurrence with teacher-selected support. Diffusion extends this support into multi-hop targets (Coifman & Lafon, 2006); mass-aware sampling keeps the target compact; and support-mass calibration aligns its probability against the unselected complement. Candidate pooling preserves the encoding reuse that makes mini-batches efficient.

Global geometry as relational knowledge $\longrightarrow$ batch-selected support $\longrightarrow$ teacher-selected support

Figure 1: **The relation-selection bottleneck.** Corpus relations are defined by teacher geometry, whereas conventional relational losses obtain support from mini-batch co-occurrence. RIPPLE instead makes support teacher-dependent while retaining compact updates.

Across multiple datasets and teacher–student settings, RIPPLE consistently improves the mean compact-student result. The gains are strongest on semantic textual similarity, whereas pair-classification results are less uniform. We therefore complement the aggregate benchmark with matched-compute comparisons and diagnostics that separate corpus exposure, teacher-selected support, diffusion, mass calibration, and candidate efficiency.

Our contributions are:

- We formulate global relational transfer as a support-selection problem, characterize the target-agnostic exposure of mini-batches, and derive a finite-epoch bound connecting RIPPLE's selected support to global geometry distortion.
- We introduce RIPPLE, which decouples teacher-driven relation selection from mini-batch computation using a cached teacher graph, multi-hop targets, mass-aware candidates, and support-mass calibration.
- We evaluate RIPPLE across diverse embedding tasks and teacher–student settings, with controlled analyses of support selection, multi-hop targets, mass calibration, and candidate efficiency.

2 Related Work

Text embedding distillation. Knowledge distillation transfers behavior from a large teacher to a compact student (Hinton et al., 2015). Language-model compression commonly matches output distributions, hidden states, or attention maps (Sanh et al., 2019; Jiao et al., 2020; Wang et al., 2020; Xu et al., 2024). Embedding-specific methods instead distill quantities used by similarity-based applications. DistilCSE combines teacher supervision with contrastive learning (Gao et al., 2021a; Xu et al., 2023); EmbedDistill transfers relative query–document geometry (Kim et al., 2023); and recent systems align embeddings, layers, tokens, or heterogeneous representation spaces (Lee et al., 2024; Zhang et al., 2024a; Vujanic & Rueckstiess, 2025; Truong et al., 2025; Dao et al., 2026; Akram et al., 2026).

Relational KD and support selection. Similarity-preserving KD matches pairwise activation similarities (Tung & Mori, 2019); RKD transfers inter-example distances and angles (Park et al., 2019); and PKT aligns conditional affinity distributions (Passalis et al., 2021). Graph- and topology-based variants transfer instance edges or persistent summaries (Liu et al., 2019; Kim et al., 2024). These methods establish relational geometry as useful knowledge, but many instantiate relations using the current mini-batch; NRKD likewise mines an in-batch similarity matrix (Gou et al., 2025). Full Kernel Matrix Transfer approximates the full pairwise kernel with Nystrom landmarks (Qian et al., 2022), while BINGO groups teacher-similar instances (Xu et al., 2022). RIPPLE isolates a different question: how to let teacher relevance select anchor-specific corpus support under bounded relational compute.

Graph geometry and diffusion. Neighborhood graphs recover manifold structure in Laplacian Eigenmaps and Diffusion Maps (Belkin & Niyogi, 2003; Coifman & Lafon, 2006); heat-based constructions provide multi-scale summaries (Sun et al., 2009; Huguet et al., 2023). Geometric KD aligns heat kernels between teacher and student graphs (Yang et al., 2022), while graph-to-MLP methods distill observed graph structure into graphless models (Zhang et al., 2022; Tian et al., 2025). RIPPLE instead induces a graph from cached text embeddings solely to select and diffuse supervision for a standard text encoder; neither graph nor teacher is used at inference.

3 RIPPLE: TEACHER-SELECTED GLOBAL RELATIONAL TRANSFER

RIPPLE uses the teacher's global geometry to select informative corpus-level relations, rather than letting random mini-batch co-occurrence determine the supervision support. A cached teacher graph selects corpus relations and defines multi-hop targets. During training, each anchor receives a compact candidate set drawn from those targets; candidates are pooled across the mini-batch for encoding reuse. A support-mass loss calibrates how much student probability lies inside the selected support. Figure 1 summarizes the distinction from batch-selected relational KD.

3.1 WHY RELATION SELECTION MATTERS

Let $\mathcal{D} = \{x_i\}_{i=1}^{N}$ be an unlabeled corpus. The teacher and student produce unit-normalized embeddings t_i and s_i. For each anchor i, let p_i^T denote the teacher's relational distribution over the remaining corpus. For candidate support C, we call $p_i^T(C)$ its *exposed teacher mass* and $1 - p_i^T(C)$ its *missed teacher mass*.

Conventional relational KD constructs C from examples that happen to share a mini-batch. The following proposition isolates the resulting limitation; its proof is given in Appendix A.

Proposition 1 (Target-agnostic mini-batch exposure). *Let $\mathcal{O}_i^{(T)}$ be the examples co-occurring with anchor i over T independent random reshuffles into batches of size B. Then, for any teacher relational distribution p_i^T,*

$$\mathbb{E}[1 - p_i^T(\mathcal{O}_i^{(T)})] = \left(1 - \frac{B-1}{N-1}\right)^T. \tag{1}$$

Expected exposure is independent of teacher relevance: important and irrelevant relations have the same inclusion probability. This motivates the central design rule of RIPPLE: teacher geometry selects supervision; mini-batches only organize computation.

3.2 TEACHER-SELECTED MULTI-SCALE TARGETS

We cache teacher embeddings and construct a mutual k-nearest-neighbor graph. Mutuality removes one-directional hub edges. Retained edges are weighted by teacher cosine similarity and row-normalized:

$$P_{ij}^T = \frac{\mathbf{1}[j \in \mathcal{M}_k^T(i)] \exp(c_{ij}^T/\tau_i)}{\sum_{\ell \in \mathcal{M}_k^T(i)} \exp(c_{i\ell}^T/\tau_i)}. \tag{2}$$

Here $\mathcal{M}_k^T(i)$ is the mutual neighborhood of i. We choose the row temperature τ_i by matching a fixed target entropy; Appendix B.2 gives the implementation. The teacher is not used after this graph is cached.

Immediate neighbors capture local teacher geometry, while connected neighborhoods carry broader structure. With the lazy transition operator $\widetilde{P}^T = \frac{1}{2}(I + P^T)$, RIPPLE uses

$$g_{i,1}^T = P_i^T, \qquad g_{i,r}^T = e_i^\top (\widetilde{P}^T)^r \Big|_{\mathcal{D}\setminus\{i\}}, \quad r \in \mathcal{R} \setminus \{1\}. \tag{3}$$

The one-step row represents direct teacher neighbors; larger r summarizes corpus regions reachable from the anchor, following the diffusion-map view of manifold geometry (Coifman & Lafon, 2006). The student predicts each target with a cosine-softmax over the candidate texts rather than by constructing a student random walk.

3.3 MASS-AWARE RELATIONAL MATCHING

Full-corpus scoring would defeat compact training. For each anchor and scale, RIPPLE retains the highest-mass relations, samples the remaining target quota in proportion to teacher mass, and adds hard and uniform negatives. Candidate sets are then pooled and deduplicated across the anchor batch, so each candidate text is encoded once while retaining anchor-specific targets.

The selected-support loss matches teacher and student distributions within each anchor's candidates. Because this restricted comparison cannot reveal alternatives outside a high-mass support, RIPPLE also forms an ambient comparison from pooled hard and random negatives. Let $\mathcal{R}_0 = \{0\} \cup \mathcal{R}$, where scale 0 denotes this ambient comparison, and let $\Omega_{i,r}$ denote the corresponding candidate domain. The relational objective is

$$\mathcal{L}_{\text{rel}} = \frac{1}{|B|} \sum_{i \in B} \sum_{r \in \mathcal{R}_0} \omega_r D_{\text{KL}} \left(g^T_{i,r}|_{\Omega_{i,r}} \| g^S_{i,r}|_{\Omega_{i,r}} \right). \tag{4}$$

The analysis reports retained teacher mass and student off-support mass alongside full-corpus divergence, rather than treating restricted KL as evidence of global matching by itself.

Finite-epoch geometry approximation. Let $C_i^{(T)}$ be the union of relations selected for anchor i through epoch T, and define the bounded cosine distortion $\ell_{ij}^{(T)} = (c_{ij}^{S,(T)} - c_{ij}^T)^2/4$. Let $\epsilon_T = N^{-1} \sum_i \mathbb{E}_{j \sim p_i^T|_{C_i^{(T)}}} [\ell_{ij}^{(T)}]$ be the final student's selected-support error and $\bar{\delta}_T = N^{-1} \sum_i [1 - p_i^T(C_i^{(T)})]$ the average missed teacher mass.

Proposition 2 (Finite-epoch global geometry approximation)**.** *The teacher-weighted global geometry distortion satisfies*

$$\begin{aligned} \mathcal{E}_T &= \frac{1}{N} \sum_i \sum_{j \neq i} p_i^T(j) \ell_{ij}^{(T)} \leq \epsilon_T + \bar{\delta}_T, \\ \mathbb{E}[\bar{\delta}_T] &\leq \frac{1}{N} \sum_i \sum_{j \neq i} p_i^T(j)(1 - p_i^T(j))^{mT}, \end{aligned} \tag{5}$$

provided RIPPLE includes at least m teacher-proportional draws per anchor and epoch. Deterministic high-mass retention and additional negatives can only reduce missed mass.

Proposition 2 separates optimization error from support error. When teacher mass is concentrated on h relations and relational compute is matched, the RIPPLE residual decays on the scale of h, whereas Proposition 1 gives corpus-size dependence for mini-batch support. RIPPLE therefore has the tighter finite-epoch upper bound whenever its selected-support optimization error does not exceed the resulting coverage advantage.

3.4 Support-Mass Calibration and Training

Conditional matching inside C_i does not determine the total student probability assigned to that support. For scale r, define

$$\alpha^T_{i,r} = g^T_{i,r}(C_i), \qquad \alpha^S_{i,r} = g^S_{i,r}(C_i). \tag{6}$$

The KL chain rule for the partition $\{C_i, \bar{C}_i\}$ gives the exact decomposition

$$\begin{aligned} D_{\text{KL}}(g^T_{i,r} \| g^S_{i,r}) = {} & D_{\text{KL}}\big(\text{Bern}(\alpha^T_{i,r}) \big\| \text{Bern}(\alpha^S_{i,r})\big) \\ & + \alpha^T_{i,r} D_{\text{KL}}(g^T_{i,r}(\cdot|C_i) \| g^S_{i,r}(\cdot|C_i)) \\ & + (1 - \alpha^T_{i,r}) D_{\text{KL}}(g^T_{i,r}(\cdot|\bar{C}_i) \| g^S_{i,r}(\cdot|\bar{C}_i)). \end{aligned} \tag{7}$$

RIPPLE directly controls the first term with

$$\mathcal{L}_{\text{mass}} = \frac{1}{|B|} \sum_{i \in B} \sum_{r \in \mathcal{R}} \omega_r D_{\text{KL}}\big(\text{Bern}(\alpha^T_{i,r}) \big\| \text{Bern}(\widehat{\alpha}^S_{i,r})\big). \tag{8}$$

Full-corpus student normalization is avoided with stratified ambient sampling. Hard negatives and the remaining corpus form disjoint strata. If $\pi_i(j)$ is an ambient item's marginal inclusion probability, then

$$\widehat{Z}^S_{i,\bar{C},r} = \sum_{j \in A_i} \frac{\exp(c^S_{ij}/\tau_{i,r})}{\pi_i(j)}, \qquad \widehat{\alpha}^S_{i,r} = \frac{Z^S_{i,C,r}}{Z^S_{i,C,r} + \widehat{Z}^S_{i,\bar{C},r}}. \tag{9}$$

The complement partition estimator is Horvitz–Thompson; the selected-support partition is scored exactly.

The complete training objective is

$$\mathcal{L}_{\mathrm{RIPPLE}} = \mathcal{L}_{\mathrm{rel}} + \lambda_{\mathrm{mass}}\mathcal{L}_{\mathrm{mass}}. \tag{10}$$

Graph construction is offline and stores only sparse neighborhoods. Each update encodes the anchor batch and its deduplicated candidate pool. At inference, deployment uses only the distilled student encoder.

4 EXPERIMENTS

We first evaluate whether RIPPLE improves compact text embeddings across teacher–student pairs and downstream tasks. Section 5 then tests the mechanisms proposed in Section 3.

4.1 EXPERIMENTAL SETUP

Models and distillation data. We evaluate three teacher→student pairs: Qwen3-Embedding-4B → BERT-base (109M), BGE-M3 → MiniLMv2-H768 (66M), and Qwen3-Embedding-0.6B → MiniLMv2-H384 (22M). The teacher families are described by Zhang et al. (2025) and Chen et al. (2024). Following the TALAS protocol (Dao et al., 2026), the unlabeled distillation corpus contains approximately 15K sentences sampled evenly from Emotion, WiC, and STS-B. Teacher embeddings are cached once; training uses neither task labels nor online teacher inference.

Evaluation. The nine downstream datasets cover classification (Banking77, Tweet, Emotion), pair classification (MRPC, SciTail, WiC), and semantic textual similarity (SICK, STS12, STS-B). We report macro-F1, average precision, and Spearman correlation, respectively. Emotion, WiC, and STS-B are in-domain; the other six datasets are out-of-domain. "Avg." is the unweighted mean of the nine primary metrics. Checkpoints and pair-classification thresholds are selected on validation data, and test data are evaluated once.

Baselines and implementation. We compare with the teacher, the undistilled student, SimCSE-unsup (Gao et al., 2021b), CDM (Chen et al., 2025), DSKD (Zhang et al., 2024b), Jasper–Stella (Zhang et al., 2024a), DistillCSE (Gao et al., 2021a; Xu et al., 2023), EMO (Truong et al., 2025), and TALAS (Dao et al., 2026). We use one canonical RIPPLE configuration across downstream tasks and vary only the component named in each controlled comparison. Appendix B gives the implementation details.

4.2 MAIN RESULTS

Table 1 reports all nine datasets for the three teacher–student settings. RIPPLE obtains the best mean compact-student score in each setting: 78.09, 77.99, and 74.91. The corresponding margins over the strongest baseline are 0.05, 1.35, and 1.45 points; the first is small relative to run variation. The most consistent gains occur on semantic textual similarity. RIPPLE gives the best compact-student STS-B result in all three settings and leads all three STS datasets for both MiniLMv2 students. Pair classification is less uniform, including lower WiC scores. The table therefore supports RIPPLE's overall competitiveness, particularly on similarity-sensitive tasks, but does not by itself identify which mechanism produces the gains.

5 ANALYSIS

We organize the analysis in the same causal order as the method: support coverage, teacher-selected adjacency, multi-hop targets, support-mass calibration, and candidate efficiency. Aggregate downstream changes are interpreted together with full-corpus relational diagnostics; they are not used alone as evidence for a mechanism.

5.1 BATCH-SELECTED TEACHER-MASS EXPOSURE

To test Proposition 1, we track exposed teacher mass against relation opportunities for batch- and teacher-selected support. During batch-local training, we compare in-batch KL with full-corpus

Table 1: Results across three teacher → student settings on nine datasets. Emotion, WiC, and STS-B (marked with ⋆) are in-domain; the remaining datasets are out-of-domain. Bold and underlined values denote the best and second-best results within each setting, excluding the Teacher and Student base reference rows. TALAS and RIPPLE results are reported as mean ± standard deviation; the remaining baseline results are from TALAS (Dao et al., 2026). For setting (c), the epoch checkpoint is selected on the validation average.

	Classification (F1)			**Pair Classification (AP)**			**STS (Spearman)**			**Domain Avg**		**Avg.**
	Banking77	Tweet	Emotion⋆	MRPC	SciTail	WiC⋆	SICK	STS12	STS-B⋆	Avg-In	Avg-Out	Avg.
(a) Qwen3-Embedding-4B → BERT-base (109M)												
Teacher	93.63	72.22	68.94	84.79	91.77	66.80	83.43	82.55	86.87	74.20	84.73	81.22
Student base	84.86	47.79	68.02	77.36	67.74	59.22	42.43	21.54	20.30	42.44	60.33	54.36
SimCSE-unsup	89.21	69.13	53.55	82.86	76.59	**71.64**	68.19	61.68	70.36	65.18	74.61	71.47
CDM	89.75	70.58	53.11	84.83	75.19	71.06	69.37	68.48	73.92	66.03	76.37	72.92
DSKD	89.66	69.93	54.51	83.13	77.95	71.53	68.65	63.37	71.42	65.82	75.45	72.24
Jasper and Stella	89.71	73.83	65.98	85.33	80.99	70.50	73.96	67.90	75.65	70.71	78.62	75.98
DistillCSE	90.94	70.90	60.80	82.86	78.98	68.63	75.12	72.61	77.08	68.84	78.57	75.32
EMO	90.78	73.54	67.48	84.46	81.57	69.76	75.66	68.83	77.17	71.47	79.14	76.58
TALAS	90.44 ± 0.13	$\mathbf{74.62 \pm 0.17}$	$\mathbf{70.87 \pm 0.52}$	$\mathbf{85.90 \pm 0.10}$	81.78 ± 0.29	68.84 ± 0.15	$\mathbf{77.56 \pm 0.10}$	73.12 ± 0.30	79.24 ± 0.10	$\mathbf{72.98 \pm 0.19}$	80.57 ± 0.01	78.04 ± 0.07
RIPPLE	$\mathbf{91.75 \pm 0.17}$	73.49 ± 0.45	70.21 ± 0.57	83.38 ± 0.05	$\mathbf{83.47 \pm 0.18}$	66.58 ± 0.06	77.21 ± 0.13	$\mathbf{75.48 \pm 0.19}$	$\mathbf{81.23 \pm 0.09}$	72.67 ± 0.14	$\mathbf{80.80 \pm 0.04}$	$\mathbf{78.09 \pm 0.04}$
(b) BGE-M3 → MiniLMv2-H768 (66M)												
Teacher	93.52	73.85	68.56	85.81	91.87	61.47	79.18	78.73	84.87	71.63	83.83	79.76
Student base	81.84	47.70	71.67	79.67	65.06	60.89	49.46	26.76	24.12	44.24	62.41	56.35
SimCSE-unsup	87.89	71.45	55.65	83.87	76.32	67.96	66.93	59.66	63.97	62.53	74.35	70.41
CDM	89.15	70.48	58.71	84.16	75.91	**70.00**	68.14	62.74	67.78	65.50	75.10	71.90
DSKD	89.51	70.53	57.27	84.39	75.36	69.96	68.81	66.78	70.05	65.76	75.90	72.52
Jasper and Stella	88.38	74.13	62.81	85.98	80.33	68.38	74.35	66.80	74.55	68.58	78.33	75.08
DistillCSE	90.50	72.70	59.88	84.39	78.48	69.49	73.97	70.00	73.73	67.70	78.34	74.79
EMO	90.19	74.26	61.48	85.31	80.82	68.30	75.41	67.06	76.35	68.71	78.84	75.46
TALAS	89.72 ± 0.19	$\mathbf{75.38 \pm 0.11}$	65.83 ± 0.58	$\mathbf{87.58 \pm 0.02}$	82.90 ± 0.12	66.96 ± 0.07	75.82 ± 0.11	68.11 ± 0.20	77.51 ± 0.18	70.10 ± 0.13	79.92 ± 0.05	76.64 ± 0.04
RIPPLE	$\mathbf{90.58 \pm 0.12}$	73.97 ± 0.30	$\mathbf{71.06 \pm 0.98}$	85.22 ± 0.09	$\mathbf{83.92 \pm 0.14}$	62.87 ± 0.15	$\mathbf{78.81 \pm 0.09}$	$\mathbf{73.99 \pm 0.20}$	$\mathbf{81.46 \pm 0.22}$	$\mathbf{71.80 \pm 0.31}$	$\mathbf{81.08 \pm 0.07}$	$\mathbf{77.99 \pm 0.13}$
(c) Qwen3-Embedding-0.6B → MiniLMv2-H384 (22M)												
Teacher	93.21	71.33	66.53	83.37	90.00	66.85	80.64	77.11	84.60	72.66	82.61	79.29
Student base	68.29	41.07	69.90	78.39	64.56	59.58	47.60	21.95	22.08	40.91	58.45	52.60
SimCSE-unsup	86.67	69.56	54.00	82.29	73.76	65.71	66.04	59.11	62.08	60.60	72.91	68.80
CDM	86.07	70.32	53.04	81.97	72.97	68.34	65.97	61.12	65.75	62.38	73.07	69.51
DSKD	85.66	69.86	52.25	82.17	73.52	68.58	66.54	62.52	66.70	62.51	73.38	69.76
Jasper and Stella	85.94	71.25	57.99	83.36	76.17	68.25	70.57	63.81	70.11	65.45	75.18	71.94
DistillCSE	**88.03**	69.87	54.24	83.20	77.66	67.88	70.09	68.19	71.53	64.55	76.17	72.30
EMO	87.03	71.67	59.63	83.45	78.17	**68.89**	70.81	65.78	71.42	66.65	76.15	72.98
TALAS	84.16 ± 0.15	$\mathbf{72.82 \pm 0.11}$	61.13 ± 0.52	$\mathbf{84.85 \pm 0.10}$	$\mathbf{80.91 \pm 0.27}$	66.32 ± 0.07	70.72 ± 0.15	67.80 ± 0.08	72.39 ± 0.31	66.61 ± 0.24	76.88 ± 0.07	73.46 ± 0.10
RIPPLE	87.50 ± 0.24	70.50 ± 0.29	$\mathbf{66.40 \pm 0.40}$	81.68 ± 0.12	80.73 ± 0.05	65.34 ± 0.10	$\mathbf{73.69 \pm 0.11}$	$\mathbf{72.08 \pm 0.09}$	$\mathbf{76.29 \pm 0.08}$	$\mathbf{69.34 \pm 0.13}$	$\mathbf{77.70 \pm 0.11}$	$\mathbf{74.91 \pm 0.08}$

JS divergence and teacher-neighbor recall. Coverage-only probes at multiple corpus sizes test the predicted dependence on corpus scale without confounding it with student optimization.

The relevant empirical question is whether teacher-selected candidates capture substantially more mass at the same number of scored relations. If an empirical teacher target is diffuse, the retained-mass curve—rather than the concentrated-support special case—determines the strength of this advantage.

5.2 Matched-Compute Support Selection

The core comparison uses batch-local KD, random corpus support, RIPPLE one-step support, RIPPLE with diffusion, full RIPPLE with mass calibration, and a degree-preserving random graph. Batch → random corpus tests broader exposure; random corpus → RIPPLE one-step tests teacher relevance at a matched number of anchor–column scores. The remaining variants isolate diffusion, support-mass calibration, and generic graph regularization. All variants share the student initialization, optimizer, schedule, seeds, and scored-pair budget; Appendix B.3 specifies the controls.

5.3 Multi-Hop Transfer and Mass Calibration

Diffusion should improve relations beyond the directly supervised one-hop band rather than merely increase neighborhood coverage. We therefore evaluate held-out diffusion errors at broader scales together with teacher–student JS in fixed teacher-rank bands. Varying only graph width tests whether additional teacher support improves these fixed bands.

For mass calibration, we compare diffusion-only training with full RIPPLE and measure selected-support mass error alongside full-corpus JS. We also compare the sampled complement-partition estimate with full scoring on a fixed diagnostic subset.

5.4 Coverage, Calibration, and Efficiency

We vary the candidate budget and measure retained teacher mass, selected-versus-unselected mass calibration, student off-support mass, sampled-to-full gradient agreement, downstream quality, step

Matched-compute analysis.

Batch-local, random corpus, RIPPLE one-step, RIPPLE diffusion,
full RIPPLE, and a degree-preserving random graph.

Downstream Avg.↑ Full-corpus JS↓ Held-out E_8 ↓.

Figure 2: **Matched-compute analysis of RIPPLE.** Random corpus support controls for broader exposure and the number of comparisons; one-step RIPPLE isolates teacher-selected adjacency; diffusion and mass-calibration variants test the remaining components; the random graph controls for generic graph regularization. Bars report mean $\pm$ standard deviation over seeds.

Mechanism and efficiency diagnostics.

(a) graph width vs. retained mass and fixed rank-band JS.
(b) candidate budget vs. calibration, gradient fidelity, time, and memory.

Figure 3: **Coverage, calibration, and efficiency.** Left: graph width is evaluated on fixed teacher-rank bands to distinguish broader transfer from trivially increased support. Right: candidate budget is evaluated through target coverage, mass calibration, gradient fidelity, downstream quality, and computational cost.

time, and peak memory. To test Proposition 2, we additionally track selected-support cosine distortion ϵ_T, missed teacher mass $\bar{\delta}_T$, their bound $\epsilon_T + \bar{\delta}_T$, and full-corpus distortion $\mathcal{E}_T$ over training epochs. High retained mass with small restricted KL is insufficient if mass calibration remains poor; these quantities are therefore reported separately. Saturation at a small candidate budget before cost approaches full scoring is the evidence required for concentration-adaptive computation.

The controlled settings are summarized in Appendix B.3. Until these mechanism diagnostics are populated, the main table supports effectiveness but not a causal attribution to relation selection, diffusion, or mass calibration.

6 Limitations and Conclusion

RIPPLE treats the teacher's global corpus geometry as a source of relational knowledge and separates the selection of that knowledge from mini-batch computation. The coverage analysis shows that random co-occurrence is independent of teacher relevance, whereas teacher-anchored selection adapts to target concentration. Empirically, RIPPLE is strongest and most consistent on semantic textual similarity; less uniform pair-classification results indicate that improved relational transfer does not benefit every task equally.

The factor-N/h separation in Appendix A assumes concentrated support, and real teacher targets may have heavy tails. Proposition 2 concerns teacher-weighted cosine distortion and remains conditional on the selected-support optimization error; it does not guarantee downstream improvement. The complement partition used for mass calibration is estimated from finite hard and uniform samples, so its variance can be substantial for highly concentrated student distributions. In addition, RIPPLE may inherit teacher errors, hubs, and domain bias. Its graph is corpus-dependent, and exact neighbor construction is quadratic before sparsification, although approximate search can replace it at larger scale.

RIPPLE nevertheless adds no inference-time machinery: deployment uses only the distilled text encoder. More broadly, the results suggest that global relational transfer does not require abandoning mini-batches; it requires preventing mini-batches from choosing the relations.

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A PROOFS FOR SUPPORT COVERAGE AND GEOMETRY APPROXIMATION

A.1 PROOF OF PROPOSITION 1

Proof. For a fixed anchor i and column $j \neq i$, a uniformly shuffled batch of size B includes j with probability $(B-1)/(N-1)$. Independence across reshuffles therefore gives

$$\Pr[j \notin \mathcal{O}_i^{(T)}] = \left(1 - \frac{B-1}{N-1}\right)^T. \tag{11}$$

Weighting this event by $p_i^T(j)$ and summing over j proves Eq. (1), because $\sum_j p_i^T(j) = 1$. $\square$

Component	Configuration
Teacher graph	Mutual kNN, $k = 200$; target perplexity 30
Diffusion	Scales $\{1, 2, 4\}$; truncation tolerance 0.01
Candidates	66 per anchor: 14 target, 26 hard, 26 random
Optimization	AdamW; batch size 32; 5 epochs; LR 2×10^{-5}
Schedule	6% warm-up; cosine decay to 3×10^{-6}
Mass calibration	Stratified hard/uniform complement; $\lambda_{\text{mass}} = 0.5$
Ambient profile	Temperature 0.10

Table 2: Canonical RIPPLE configuration. Controlled comparisons vary only the named component.

A.2 Proof of Proposition 2

Proof. For anchor i, write $C_i = C_i^{(T)}$, $\delta_i = 1 - p_i^T(C_i)$, and $\epsilon_i = \mathbb{E}_{j \sim p_i^T|_{C_i}}[\ell_{ij}^{(T)}]$. Splitting its global distortion over selected and unselected relations gives

$$\sum_{j \neq i} p_i^T(j)\ell_{ij}^{(T)} = p_i^T(C_i)\epsilon_i + \sum_{j \notin C_i} p_i^T(j)\ell_{ij}^{(T)} \tag{12}$$

$$\leq \epsilon_i + \delta_i, \tag{13}$$

because $0 \leq \ell_{ij}^{(T)} \leq 1$. Averaging over anchors proves $\mathcal{E}_T \leq \epsilon_T + \bar{\delta}_T$.

It remains to bound missed mass. A relation retained deterministically has zero miss probability. For any remaining relation j, proportional sampling without replacement selects j at each draw with conditional probability at least $p_i^T(j)$, because removing other entries can only decrease the remaining normalizer. Thus its probability of being missed by at least mT proportional draws is at most $(1 - p_i^T(j))^{mT}$. Multiplying by teacher mass, summing over relations and anchors, and taking expectation proves the second line of Eq. (5). □

When B is small relative to N, Proposition 1 requires $\Theta(N \log(1/\epsilon))$ batch relations per anchor to reduce expected missed mass below ϵ. If p_i^T is uniform on h relevant relations, the residual term in Proposition 2 is at most $(1 - 1/h)^{mT}$, requiring $\Theta(h \log(1/\epsilon))$ teacher-proportional draws. This yields the factor-$\Theta(N/h)$ separation used in the paper, but only for the concentrated-support regime.

B Implementation and Evaluation Details

B.1 Canonical Configuration

Teacher embeddings, graph neighborhoods, and diffusion targets are computed once before training. The teacher is not loaded during student optimization. All downstream tasks use the same RIPPLE configuration.

B.2 Graph, Candidates, and Complement Sampling

Entropy-calibrated bandwidth. For each graph row, we choose its softmax temperature by bisection so that the transition distribution has the target perplexity in Table 2. Rows whose degree is below the target use the largest feasible entropy; degree-one and constant-score rows use the uniform distribution on their available support. Broader diffusion profiles use temperature scaled by the square root of the walk scale and loss weight inversely proportional to scale. Sparse profiles retain the smallest support carrying at least 99% of their mass.

For candidate selection, RIPPLE keeps the highest-mass target entries and samples the remaining target quota without replacement in proportion to teacher mass. The complement is partitioned into a hard-negative stratum and all remaining corpus items. Each stratum is sampled uniformly without replacement using the quotas in Table 2; every sampled ambient item carries inverse marginal inclusion probability H/k_h or U/k_u. Candidates are deduplicated across anchors before encoding.

B.3 Controlled Comparisons

Core ablations. All variants use the same initialization, optimizer, epochs, seeds, and scored anchor–column budget. Batch-local KD normalizes over co-occurring examples; random corpus support uses a uniformly sampled support of matched size; one-step RIPPLE uses only direct graph rows; diffusion adds broader targets; and full RIPPLE adds support-mass calibration. The random-graph control preserves graph size and degree sequence while rewiring adjacency, then uses the same targets, sampler, and loss construction.

Mechanism and efficiency diagnostics. Graph-width comparisons hold candidate budget and optimization fixed and report retained teacher mass plus JS divergence in fixed teacher-rank bands. Candidate-budget comparisons report retained mass, selected-versus-unselected mass calibration, student off-support mass, gradient agreement with full scoring, time, and peak memory. Mass-calibration comparisons report Bernoulli KL, absolute support-mass error, and sampled-versus-full complement-partition error. All diagnostics use the same anchor set, hardware, and software stack across variants.